B.Tech. (Computer Engg.) V Semester Examinations 20BCS021

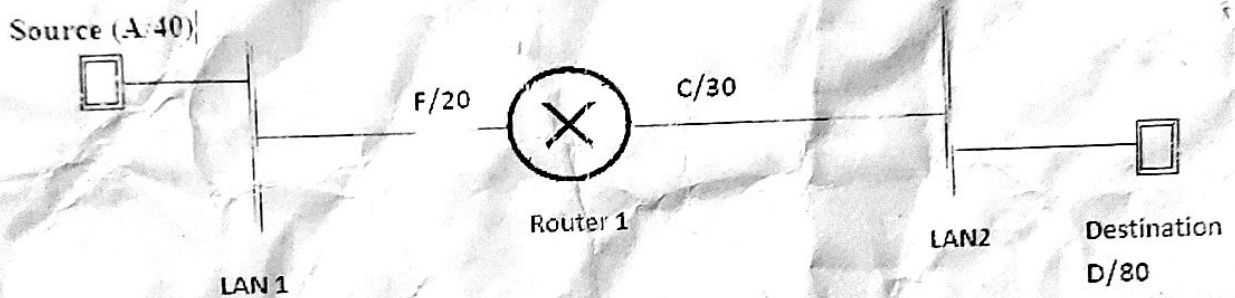
Session 2022-23

Time: 1 hr. **Computer Networks (CEN - 505) Sessional-I** Max. Marks: 15

Note: Attempt All Questions

1. CO1 In TCP/IP protocol suite, explain how can transport layer be both connection-oriented and connectionless, when the underlying network layer protocol is connectionless only?

2. CO1 In the following figure, device with IP address 'A' sends a message to device with IP address 'D' via LAN1, router1 and LAN2. Assume that the communication is between a process running at system 'A' with port address 'i' and a process running at system 'D' with port address 'j'. Discuss the complete scenario for this transmission and show the contents of packets and frames at the network layer, data link layer and transport layer at each hop.



3. CO1 An organization requires a range of IP addresses to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one to the organization?

I. 202.61.84.0/21
II. 202.61.104.0/21
III. 202.61.64.0/21
IV. 202.61.144.0/21

(a) I and IV only (b) III and IV only (c) I and II only (d) II and III only

Justify your answer.

4. CO2 What are the major problems in NRZ line coding scheme and how can it be handled with the bipolar scheme? Sketch the HDB3 and 2B1Q coding for the following bit stream, assume that the number of 1's so far are odd and last signal was at low level:

111000001001000010000101